

Reasoning Behavior Analysis Report

EXPERIMENTAL SETUP

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Data Folder: data/MATH500/deepseek/p2
Total Items: 500
Total Strategies: 9

OVERALL ACCURACY

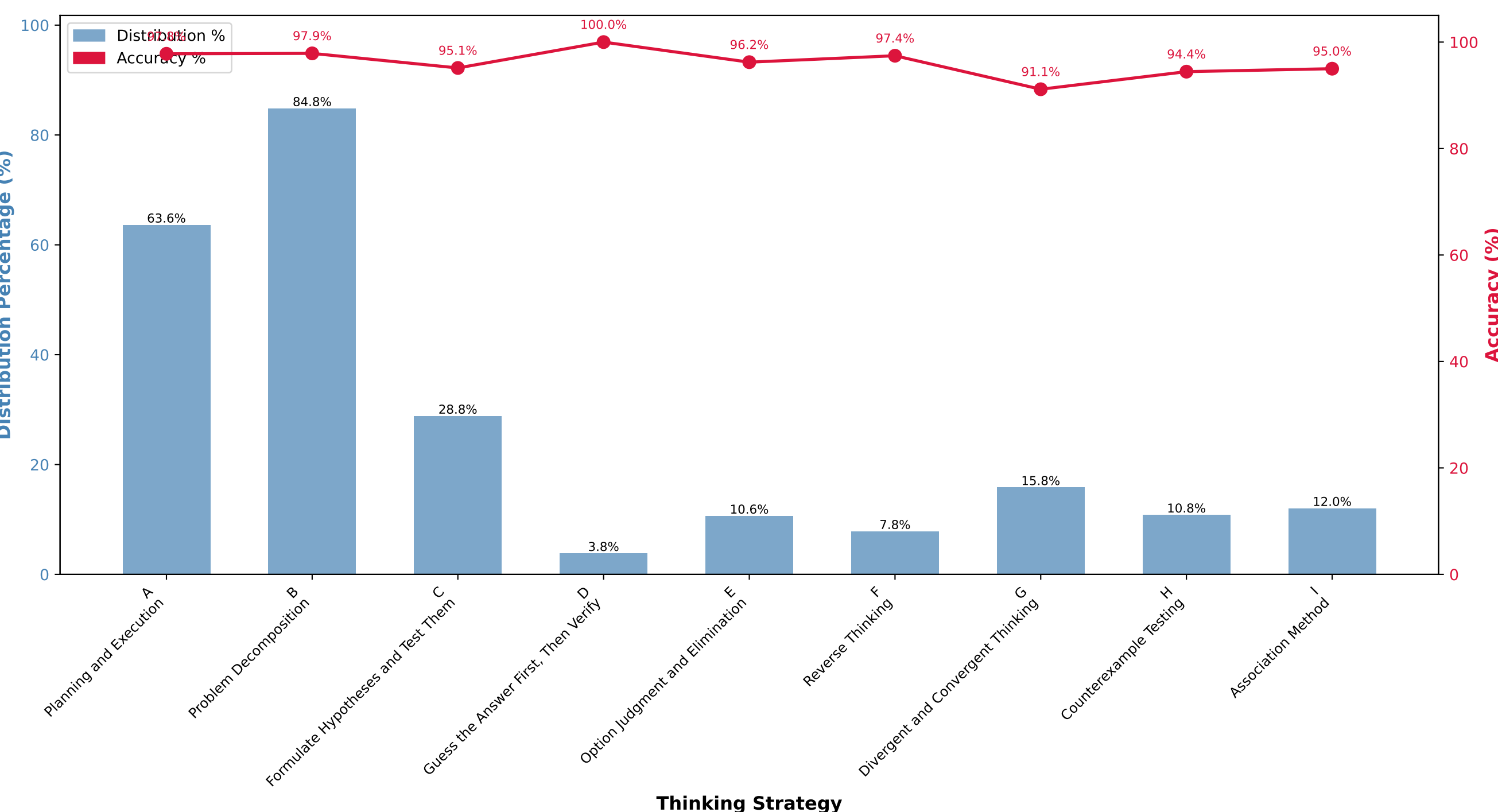
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Total Problems: 500
Correct: 491
Incorrect: 9
Overall Accuracy: 98.20%

THINKING STRATEGY DISTRIBUTION

- =====
- A: Planning and Execution
Distribution: 63.6%
Accuracy: 97.80% (311/318)
 - B: Problem Decomposition
Distribution: 84.8%
Accuracy: 97.88% (415/424)
 - C: Formulate Hypotheses and Test Them
Distribution: 28.8%
Accuracy: 95.14% (137/144)
 - D: Guess the Answer First, Then Verify
Distribution: 3.8%
Accuracy: 100.00% (19/19)
 - E: Option Judgment and Elimination
Distribution: 10.6%
Accuracy: 96.23% (51/53)
 - F: Reverse Thinking
Distribution: 7.8%
Accuracy: 97.44% (38/39)
 - G: Divergent and Convergent Thinking
Distribution: 15.8%
Accuracy: 91.14% (72/79)
 - H: Counterexample Testing
Distribution: 10.8%
Accuracy: 94.44% (51/54)
 - I: Association Method
Distribution: 12.0%
Accuracy: 95.00% (57/60)

Thinking Strategy Distribution and Accuracy



Example for Strategy A: Planning and Execution

PROBLEM

Let f be defined by $f(x) = \begin{cases} 3-x & \text{if } x \leq 3, \\ -x^3+2x^2+3x & \text{if } x > 3. \end{cases}$
Calculate $f^{-1}(0)+f^{-1}(6)$.

GROUND TRUTH ANSWER

0

MODEL RESPONSE

The function f is defined piecewise as:

$f(x) = \begin{cases} 3-x & \text{if } x \leq 3, \\ -x^3+2x^2+3x & \text{if } x > 3. \end{cases}$

To find $f^{-1}(0)$, solve $f(x) = 0$:

- For $x \leq 3$, $3-x = 0$ implies $x = 3$.
 - For $x > 3$, $-x^3+2x^2+3x = 0$ has solutions $x = 0, 3, -1$, but none satisfy $x > 3$.
- Thus, $f^{-1}(0) = 3$.

To find $f^{-1}(6)$, solve $f(x) = 6$:

- For $x \leq 3$, $3-x = 6$ implies $x = -3$.
 - For $x > 3$, $-x^3+2x^2+3x = 6$ simplifies to $x^3-2x^2-3x+6 = 0$, which has roots $x = 2, \sqrt{3}, -\sqrt{3}$, but none satisfy $x > 3$.
- Thus, $f^{-1}(6) = -3$.

Therefore, $f^{-1}(0) + f^{-1}(6) = 3 + (-3) = 0$.

$\boxed{0}$

CHAIN OF THOUGHT (First 1500 chars)

First, I need to find $f^{-1}(0) + f^{-1}(6)$. That means I have to find the inverse of function f at 0 and at 6, and then add them together. But f is piecewise, so I need to be careful about which piece to use for the inverse.

The function f is defined as:

- $f(x) = 3 - x$ for $x \leq 3$
- $f(x) = -x^3 + 2x^2 + 3x$ for $x > 3$

To find $f^{-1}(y)$ for some y , I need to find x such that $f(x) = y$. Since f is piecewise, I might have to consider both pieces separately for each y .

Let me start with $f^{-1}(0)$. So I need x such that $f(x) = 0$.

Case 1: If $x \leq 3$, $f(x) = 3 - x = 0 \Rightarrow x = 3$.

Case 2: If $x > 3$, $f(x) = -x^3 + 2x^2 + 3x = 0$.

Let me solve $-x^3 + 2x^2 + 3x = 0$.

Factor out $-x$: $-x(x^2 - 2x - 3) = 0$.

So $-x = 0$ or $x^2 - 2x - 3 = 0$.

$-x=0 \Rightarrow x=0$, but $x=0$ is not greater than 3, so for $x>3$, this doesn't apply.

$x^2 - 2x - 3 = 0 \Rightarrow (x-3)(x+1)=0 \Rightarrow x=3$ or $x=-1$.

But $x=3$ is not greater than 3 (since $x>3$), and $x=-1$ is not greater than 3. So for $x>3$, there is no solution to $f(x)=0$.

Therefore, only $x=3$ gives $f(x)=0$, and since $x=3$ is included in the first piece, $f^{-1}(0) = 3$.

Now, $f^{-1}(6)$. I need x such that $f(x)=6$.

Again, consider both pieces.

First, for $x \leq 3$: $f(x) = 3 - x = 6 \Rightarrow 3 - x = 6 \Rightarrow -x = 3 \Rightarrow x = -3$.

And $x=-3$ is less than 3, so this is valid.

Now, for $x > 3$: $f(x) = -x^3 + 2x^2 + 3x = 6$.

So I need to solve $-x^3 + 2x^2 + 3x - 6 = 0$.

Multiply both sides by -1 : $x^3 - 2x^2 - 3x + 6 = 0$.

Now, let me factor this cubic equation.

Try to find roots. Possible rational roots are fa...

Example for Strategy B: Problem Decomposition

PROBLEM

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$\text{\textbf{Juan's Old Stamping Grounds}}$

Juan organizes the stamps in his collection by country and by the decade in which they were issued. The prices he paid for them at a stamp shop were: Brazil and France, \$6\$ cents each, Peru \$4\$ cents each, and Spain \$5\$ cents each. (Brazil and Peru are South American countries and France and Spain are in Europe.) [asy]

```
/* AMC8 2002 #8, 9, 10 Problem */
size(3inch, 1.5inch);
for ( int y = 0; y <= 5; ++y )
{
```

```
draw((0,y)--(18,y));
}
draw((0,0)--(0,5)...
```

GROUND TRUTH ANSWER

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5.4 $\text{\text{cents}}$

MODEL RESPONSE

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The average price of Juan's stamps from the 1970s is calculated as follows:

- Brazil: 12 stamps $\times$ 6 cents = 72 cents
- France: 12 stamps $\times$ 6 cents = 72 cents
- Peru: 6 stamps $\times$ 4 cents = 24 cents
- Spain: 13 stamps $\times$ 5 cents = 65 cents

Total cost for 1970s stamps: $72 + 72 + 24 + 65 = 233$ cents

Total number of 1970s stamps: $12 + 12 + 6 + 13 = 43$ stamps

Average price: $233 / 43 \approx 5.4186$ cents

Rounded to the nearest tenth of a cent: 5.4 cents

$\boxed{5.4}$

CHAIN OF THOUGHT (First 1500 chars)

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First, I need to find the average price of Juan's stamps from the 1970s. The table shows the number of stamps from each country in each decade. For the 70s, I see:

- Brazil: 12 stamps
- France: 12 stamps
- Peru: 6 stamps
- Spain: 13 stamps

The prices are:

- Brazil: 6 cents each
- France: 6 cents each (since it says Brazil and France are both 6 cents)
- Peru: 4 cents each
- Spain: 5 cents each

Now, for the 70s stamps, I need to find the total cost and then divide by the total number of stamps to get the average.

Calculate the total cost for 70s stamps:

- Brazil: 12 stamps $\times$ 6 cents = 72 cents
- France: 12 stamps $\times$ 6 cents = 72 cents
- Peru: 6 stamps $\times$ 4 cents = 24 cents
- Spain: 13 stamps $\times$ 5 cents = 65 cents

Now, sum these up: $72 + 72 = 144$; $144 + 24 = 168$; $168 + 65 = 233$ cents total cost.

Total number of 70s stamps: $12 + 12 + 6 + 13$. Let's add: $12+12=24$; $24+6=30$; $30+13=43$ stamps.

So, average price = total cost / total number = $233 / 43$.

Now, I need to calculate that and round to the nearest tenth of a cent.

Let me divide 233 by 43.

43 times 5 is 215, so $233 - 215 = 18$. So, $5 + 18/43$.

$18/43$ is approximately 0.4186, so about 5.4186 cents.

Now, rounding to the nearest tenth: look at the hundredths place, which is 1, so it rounds down. So, 5.4 cents? But let me confirm.

$233 \div 43$.

Let me do the division properly.

$43 \times 5.4 = 43 \times 5 = 215$, $43 \times 0.4 = 17.2$, so $215 + 17.2 = 232.2$, which is less than 233.

$233 - 232.2 = 0.8$, so $5.4 + 0.8/43 \approx 5.4 + 0.0186 \approx 5.4186$, as ...

Example for Strategy C: Formulate Hypotheses and Test Them

PROBLEM

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If $re^{i\theta}$ is a root of $z^8 - z^7 + z^6 - z^5 + z^4 - z^3 + z^2 - z + 1 = 0$, where $r > 0$ and $0 \leq \theta < 2\pi$, then find the sum of all possible values of θ .

GROUND TRUTH ANSWER

=====
 8π

MODEL RESPONSE

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The given polynomial is $z^8 - z^7 + z^6 - z^5 + z^4 - z^3 + z^2 - z + 1 = 0$. This can be expressed as a geometric series:
 $1 - z + z^2 - z^3 + z^4 - z^5 + z^6 - z^7 + z^8 = \frac{1 + z^9}{1 + z}$,
for $z \neq -1$. The roots of the polynomial are the roots of $1 + z^9 = 0$ excluding $z = -1$, since $z = -1$ is not a root of the polynomial.

The equation $z^9 = -1$ has solutions given by:
 $z = e^{i(2k\pi + \pi)/9} = e^{i\pi(2k+1)/9}$, $\text{quad } k = 0, 1, 2, \dots, 8$.
Thus, the arguments are $\theta_k = \frac{\pi(2k+1)}{9}$ for $k = 0$ to 8 .

The root $z = -1$ corresponds to $\theta = \pi$, which occurs when $k = 4$. This root is excluded from the polynomial's roots. Therefore, the possible values of θ are...

CHAIN OF THOUGHT (First 1500 chars)

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I need to find the sum of all possible values of θ for the roots of this polynomial. The polynomial is $z^8 - z^7 + z^6 - z^5 + z^4 - z^3 + z^2 - z + 1 = 0$. The roots are given in polar form as $re^{i\theta}$, and I need to find the sum of all θ where $r > 0$ and $0 \leq \theta < 2\pi$.

First, I notice that this polynomial looks like a geometric series. Let me write it as the sum of terms from z^8 down to z^0 , but with alternating signs. Actually, it's $z^8 - z^7 + z^6 - z^5 + z^4 - z^3 + z^2 - z + 1$. So, it's the sum from $k=0$ to 8 of $(-1)^k z^{8-k}$? Let's see: for $k=0$, $(-1)^0 z^8 = z^8$, good; $k=1$, $(-1)^1 z^7 = -z^7$, yes; but it goes to z^0 , which is 1, so for $k=8$, $(-1)^8 z^0 = 1$, perfect. So, it's sum from $k=0$ to 8 of $(-1)^k z^{8-k}$. But that seems messy.

I recall that for geometric series, sum from $n=0$ to N of $r^n = (1 - r^{N+1})/(1 - r)$. Here, it's alternating, so it might be similar to that.

Consider the sum $S = 1 - z + z^2 - z^3 + \dots$ up to z^8 . But the polynomial has z^8 first, so let's write it in descending order or ascending.

The polynomial is $z^8 - z^7 + z^6 - z^5 + z^4 - z^3 + z^2 - z + 1$. If I factor out z^8 , it might not help. Notice that this is almost the sum of a geometric series with first term 1 and ratio $-z$.

Let me define $S = 1 - z + z^2 - z^3 + z^4 - z^5 + z^6 - z^7 + z^8$. But in the polynomial, it's $z^8 - z^7 + \dots + 1$, which is the same as S since addition is commutative. So, $S = 1 - z + z^2 - z^3 + z^4 - z^5 + z^6 - z^7 + z^8$.

This is a geometric series with first term $a=1$, common ratio $r=-z$, and number of terms $n=9$. So, ...

Example for Strategy D: Guess the Answer First, Then Verify

PROBLEM

Let $a, b, c,$ and d be positive real numbers such that $a + b + c + d = 10$. Find the maximum value of $ab^2 c^3 d^4$.

GROUND TRUTH ANSWER

27648

MODEL RESPONSE

The maximum value of $(ab^2 c^3 d^4)$ subject to $(a + b + c + d = 10)$ with $(a, b, c, d > 0)$ is achieved when the variables are proportional to their exponents. That is, $(a : b : c : d = 1 : 2 : 3 : 4)$.

Given the sum $(a + b + c + d = 10)$, we have:

$a = 1, \quad b = 2, \quad c = 3, \quad d = 4.$

The product is:

$ab^2 c^3 d^4 = 1 \cdot 2^2 \cdot 3^3 \cdot 4^4 = 1 \cdot 4 \cdot 27 \cdot 256.$

Calculating step by step:

$4 \cdot 27 = 108, \quad 108 \cdot 256 = 27648.$

Thus, the maximum value is 27648.

$\boxed{27648}$

CHAIN OF THOUGHT (First 1500 chars)

First, I need to maximize $(ab^2 c^3 d^4)$ given that $(a + b + c + d = 10)$ with all positive real numbers. This looks like an optimization problem with constraints, so I should use the method of Lagrange multipliers or perhaps an inequality like AM-GM.

I recall that for products like this, the maximum often occurs when the variables are proportional to their exponents. Let me think about the exponents: a has exponent 1, b has 2, c has 3, d has 4. So the sum of exponents is $1+2+3+4=10$. That's interesting because $a+b+c+d=10$, so the sum of the variables equals the sum of the exponents. That might be a hint.

I remember that for fixed sum, the product $(x_1^{p_1} x_2^{p_2} \cdots x_n^{p_n})$ is maximized when the variables are proportional to their exponents. So here, we have $(a^1 b^2 c^3 d^4)$, so we should set $(a : b : c : d = 1 : 2 : 3 : 4)$.

Let me verify that. The sum of the ratios is $1+2+3+4=10$, which matches the sum. So, let $(a = k \cdot 1)$, $(b = k \cdot 2)$, $(c = k \cdot 3)$, $(d = k \cdot 4)$, but since the sum is 10, we have $k(1+2+3+4)=10k=10$, so $k=1$. Therefore, $a=1, b=2, c=3, d=4$.

Then the product is $(a b^2 c^3 d^4 = 1 \cdot 2^2 \cdot 3^3 \cdot 4^4)$. Let me calculate that.

First, $(2^2 = 4)$, $(3^3 = 27)$, $(4^4 = 256)$. So, $1 \cdot 4 \cdot 27 \cdot 256$. Let's compute step by step: $4 \cdot 27 = 108$, then $108 \cdot 256 = 27648$. So product is 27648.

But is this the maxim...

Example for Strategy E: Option Judgment and Elimination

PROBLEM

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If $\mathbf{a}$ and $\mathbf{b}$ are vectors such that $|\mathbf{a}| = 2$, $|\mathbf{b}| = 7$, and $|\mathbf{a} \times \mathbf{b}| = \begin{vmatrix} 3 & 2 & 6 \end{vmatrix}$, then find the smallest possible angle between $\mathbf{a}$ and $\mathbf{b}$, in degrees.

GROUND TRUTH ANSWER

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30°

MODEL RESPONSE

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The magnitude of the cross product $|\mathbf{a} \times \mathbf{b}|$ is given by $|\mathbf{a}| |\mathbf{b}| \sin \theta$, where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$.

Given $|\mathbf{a}| = 2$, $|\mathbf{b}| = 7$, and $|\mathbf{a} \times \mathbf{b}| = \begin{vmatrix} 3 & 2 & 6 \end{vmatrix}$, the magnitude of the cross product is:

$$|\mathbf{a} \times \mathbf{b}| = \sqrt{3^2 + 2^2 + 6^2} = \sqrt{9 + 4 + 36} = \sqrt{49} = 7.$$

Substitute into the formula:

$$\begin{aligned} 7 &= 2 \times 7 \times \sin \theta \\ 7 &= 14 \sin \theta \\ \sin \theta &= \frac{7}{14} = \frac{1}{2} \end{aligned}$$

Thus, $\theta = \sin^{-1} \left(\frac{1}{2} \right)$, which gives $\theta = \dots$

CHAIN OF THOUGHT (First 1500 chars)

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I need to find the smallest possible angle between vectors $\mathbf{a}$ and $\mathbf{b}$ given that their magnitudes are 2 and 7 respectively, and their cross product is (3, 2, 6). The angle between them should be related to the magnitude of the cross product.

I recall that the magnitude of the cross product is given by $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}| |\mathbf{b}| \sin \theta$, where θ is the angle between them. So, first, I should find the magnitude of the cross product vector.

The cross product is given as $\begin{vmatrix} 3 & 2 & 6 \end{vmatrix}$, so its magnitude is $\sqrt{3^2 + 2^2 + 6^2} = \sqrt{49} = 7$.

So, $|\mathbf{a} \times \mathbf{b}| = 7$.

Now, $|\mathbf{a}| = 2$ and $|\mathbf{b}| = 7$, so plugging into the formula:

$$7 = 2 \times 7 \times \sin \theta$$

That simplifies to:

$$7 = 14 \sin \theta$$

So, $\sin \theta = \frac{7}{14} = \frac{1}{2}$.

Therefore, $\theta = \sin^{-1} \left(\frac{1}{2} \right)$, which is 30 degrees or 150 degrees. But the question asks for the smallest possible angle, so it should be 30 degrees.

I feel like I might be missing something because the problem says "the smallest possible angle," which suggests that there might be multiple angles, but from the cross product magnitude, we have a fixed sine value, so the angle should be either 30 or 150, and the smallest is 30.

But let me double-check. Is there any constraint...

Example for Strategy F: Reverse Thinking

PROBLEM

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A worker receives an annual wage of $\$20,000$, which he always deposits into a savings account at the end of the year. By the end of the third year (when he makes the third deposit), he wants to have at least $\$66,200$ in the account to finance the purchase of a house. What is the minimal compound interest rate that the savings account must provide? Express your answer as a percentage, but do not include the percent sign.

GROUND TRUTH ANSWER

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10

MODEL RESPONSE

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10

CHAIN OF THOUGHT (First 1500 chars)

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First, the worker deposits $\$20,000$ at the end of each year. He makes three deposits: one at the end of year 1, one at end of year 2, and one at end of year 3. By the end of year 3, after the third deposit, he wants at least $\$66,200$.

I need to find the minimal compound interest rate that will make this happen. Since it's compound interest, and deposits are made annually, I should consider the future value of each deposit.

Let the annual interest rate be r (as a decimal). So, for example, if r is 5%, then $r = 0.05$.

Now, let's think about the future value of each deposit at the end of year 3.

- The first deposit is made at the end of year 1. It will earn interest for 2 years (from end of year 1 to end of year 3). So, its future value is $20,000 * (1 + r)^2$.

- The second deposit is made at the end of year 2. It will earn interest for 1 year (from end of year 2 to end of year 3). So, its future value is $20,000 * (1 + r)^1$.

- The third deposit is made at the end of year 3. It doesn't earn any interest yet because it's deposited at the end, so its value is just $20,000$.

Therefore, the total amount at the end of year 3 is the sum of these three future values:

$$A = 20,000 * (1 + r)^2 + 20,000 * (1 + r) + 20,000$$

We want this amount to be at least 66,200. Since we need the minimal rate, we can set it equal to 66,200 for the minimum.

So:

$$20,000 * (1 + r)^2 + 20,000 * (1 + r) + 20,000 = 66,200$$

I can factor out 20,000:

$$20,000 * [(1 + r)^2 + (1 + r) + 1] = 66,200$$

Now, divide ...

Example for Strategy G: Divergent and Convergent Thinking

PROBLEM

Two numbers, x and y are selected at random from the interval $(0,3)$. What is the probability that a triangle with sides of length 1, x , and y exists?

GROUND TRUTH ANSWER

$$\frac{1}{2}$$

MODEL RESPONSE

The probability that a triangle with sides of length 1, x , and y exists is found by considering the conditions for triangle inequality: $(1 + x > y)$, $(1 + y > x)$, and $(x + y > 1)$. These conditions simplify to $(y > |x - 1|)$ and $(y < 1 + x)$, with $(x, y \in (0, 3))$.

The sample space is the square $((0, 3) \times (0, 3))$ with area (9) . The area where the triangle exists is calculated by integrating the valid range of (y) for each (x) from (0) to (3) :

- For $(x \in (0, 1])$, the length of (y) is $(2x)$.
- For $(x \in [1, 2])$, the length of (y) is (2) .
- For $(x \in [2, 3])$, the length of (y) is $(4 - x)$.

The area is:

$$\int_0^1 2x \, dx + \int_1^2 2 \, dx + \int_2^3 (4 - x) \, dx = \dots$$

CHAIN OF THOUGHT (First 1500 chars)

First, the problem involves two numbers, x and y , selected randomly from the interval $(0,3)$. I need to find the probability that a triangle with sides of length 1, x , and y exists.

For a triangle to exist, the triangle inequality must hold. That means:

1. The sum of any two sides must be greater than the third side.

So, for sides 1, x , and y , the conditions are:

- $1 + x > y$
- $1 + y > x$
- $x + y > 1$

Since x and y are both greater than 0 (from the interval $(0,3)$), the conditions $1 + x > y$ and $1 + y > x$ might be automatically satisfied if x and y are small, but I need to consider the range.

x and y are between 0 and 3, so let's list the inequalities:

1. $1 + x > y$
2. $1 + y > x$
3. $x + y > 1$

Now, since x and y are positive, condition 3 might be the most restrictive, but I need to see when all are true.

I should consider the range of x and y . Both are from 0 to 3, so the sample space is all points (x,y) in the square $(0,3) \times (0,3)$. The area of this square is $3 * 3 = 9$.

I need to find the area within this square where the triangle inequalities hold, and then probability is that area divided by 9.

So, let's find the region where:

- a) $1 + x > y$
- b) $1 + y > x$
- c) $x + y > 1$

Since x and y are symmetric, the region might be symmetric around the line $x = y$.

Let me solve these inequalities.

First, inequality a: $1 + x > y \Rightarrow y < 1 + x$

Inequality b: $1 + y > x \Rightarrow y > x - 1$

Inequality c: $x + y > 1 \Rightarrow y > 1 - x$

Now, since x and y are between 0 and 3, I need to see which of these ...

Example for Strategy H: Counterexample Testing

PROBLEM

Consider the geometric series $4+\frac{12}{a}+\frac{36}{a^2}+\cdots$. If the sum is a perfect square, what is the smallest possible value of a where a is a positive integer?

GROUND TRUTH ANSWER

4

MODEL RESPONSE

The given geometric series is $(4 + \frac{12}{a} + \frac{36}{a^2} + \cdots)$, with first term $(u_1 = 4)$ and common ratio $(r = \frac{3}{a})$. For the series to converge, $(|r| < 1)$, which requires $(a > 3)$ since (a) is a positive integer.

The sum (S) of the series is given by:

$$S = \frac{u_1}{1 - r} = \frac{4}{1 - \frac{3}{a}} = \frac{4a}{a - 3}$$

The sum (S) is a perfect square, so $(\frac{4a}{a - 3} = k^2)$ for some integer (k) . Solving for (a) :

$$4a = k^2 (a - 3)$$

$$4a = k^2 a - 3k^2$$

$$a(4 - k^2) = -3k^2$$

$$a(k^2 - 4) = 3k^2$$

$$a = \frac{3k^2}{k^2 - 4}$$

For (a) to be an integer, $(\frac{3k^2}{k^2 - 4})$ must be an integer. This can be rewritten as:

$$a = 3 + \frac{12}{k^2 - 4}$$

where $(m = k^2 - 4)$.

CHAIN OF THOUGHT (First 1500 chars)

First, I need to understand the geometric series given: $(4 + \frac{12}{a} + \frac{36}{a^2} + \cdots)$. It's an infinite geometric series because it goes on forever. The sum of an infinite geometric series is given by $(S = \frac{u_1}{1 - r})$, where (u_1) is the first term and (r) is the common ratio.

Here, the first term $(u_1 = 4)$. To find the common ratio (r) , I look at the ratio between consecutive terms. The second term is $(\frac{12}{a})$, so $(r = \frac{\frac{12}{a}}{4} = \frac{12}{4a} = \frac{3}{a})$. Similarly, between the second and third term: $(\frac{\frac{36}{a^2}}{\frac{12}{a}} = \frac{36}{12a} = \frac{3}{a})$. So yes, $(r = \frac{3}{a})$.

For the series to converge, we need $(|r| < 1)$, so $(|\frac{3}{a}| < 1)$. Since a is a positive integer, $(\frac{3}{a} < 1)$ implies $(a > 3)$. So a must be at least 4 for the series to have a finite sum.

The sum S is given by:

$$S = \frac{u_1}{1 - r} = \frac{4}{1 - \frac{3}{a}} = \frac{4}{\frac{a - 3}{a}} = 4 \times \frac{a}{a - 3} = \frac{4a}{a - 3}$$

We are told that this sum is a perfect square. Let me denote that S is a perfect square, so $(\frac{4a}{a - 3} = k^2)$ for some integer k .

Since S is a sum of positive terms, k should be a positive integer.

$$\text{So, } (\frac{4a}{a - 3} = k^2)$$

I need to solve f...

Example for Strategy I: Association Method

PROBLEM

Let G and H denote the centroid and orthocenter of triangle ABC , respectively. Let F be the midpoint of $\overline{GH}$. Express $AF^2 + BF^2 + CF^2$ in terms of the side lengths a, b, c and circumradius R of triangle ABC .

GROUND TRUTH ANSWER

$3R^2$

MODEL RESPONSE

The point F is the midpoint of the segment joining the centroid G and the orthocenter H of triangle ABC . The sum of the squares of the distances from F to the vertices A, B , and C is given by:

$$AF^2 + BF^2 + CF^2 = 3R^2$$

where R is the circumradius of triangle ABC . This result is independent of the side lengths a, b , and c , as the terms involving $a^2 + b^2 + c^2$ cancel out in the derivation.

The derivation uses the vector approach with the circumcenter O as the origin, where $OH^2 = 9R^2 - a^2 - b^2 - c^2$ and $OG = OH/3$. Since F is the midpoint of GH , it follows that $FG^2 = GH^2 / 4$, and using the formula $AF^2 + BF^2 + CF^2 = 3FG^2 + (a^2 + b^2 + c^2)/3$, the expression simplifies to $3R^2$.

This r...

CHAIN OF THOUGHT (First 1500 chars)

First, I need to find an expression for $AF^2 + BF^2 + CF^2$ in terms of a, b, c , and R , where F is the midpoint of GH , with G being the centroid and H the orthocenter of triangle ABC .

I recall some properties of the centroid and orthocenter. The centroid G has coordinates that are the average of the coordinates of A, B , and C . The orthocenter H is the intersection of the altitudes.

I remember that in vector geometry, there are formulas for points like these. Let me denote the position vectors of A, B, C with respect to some origin O . But since we're dealing with distances, I might need to use vector identities.

I know that for any point P , $AP^2 + BP^2 + CP^2$ can be expressed in terms of the sides and the position of P relative to the centroid or something.

There's a formula: for any point P , $AP^2 + BP^2 + CP^2 = 3PG^2 + AG^2 + BG^2 + CG^2$, but I'm not sure if that helps directly.

I should find the position of F . Since F is the midpoint of GH , I can express vector OF in terms of vectors OG and OH .

Let me set O as an arbitrary origin. Then, vector $OG = (OA + OB + OC)/3$ because G is the centroid.

Now, for the orthocenter H , I recall that in vector terms, $OH = OA + OB + OC - 2OO'$, where O' is the circumcenter, but I'm not sure. I think there's a relation.

I remember that for any point P , the relation between P, G , and H might be useful.

I think there's a known fact that the centroid divides the segment from orthocenter to circumcenter in the ratio 2:1. But here we have F as midpoint o...